AI-Powered Unified Sri Lankan Tourism Platform Using GraphRAG and LLM Orchestration

Research project R26-IT-050

Abstract— Tourism planning involves decisions about attractions, accommodation, travel time and transport suitability that must remain consistent across a trip. We present CeylonGo, a research prototype combining four components: visual landmark recognition with a contextual guide, knowledge graph retrieval of hotels and activities, preference based itinerary generation, and route hazard classification with vehicle recommendation. A React Native interface connects an Express backend, Python inference services, Neo4j and itinerary persistence. The itinerary module operates on 40 source traced points of interest in Sri Lanka's Central Province and supports road network travel estimates, stop replacement and schedule rechecking. Component evaluations reveal different levels of maturity. A recorded landmark experiment reports 93.18% validation accuracy for fine tuned MobileNetV2 features with a support vector machine, but an inference feature mismatch prevents treating this as deployed application accuracy. A route grouped hazard evaluation reports 88.20% accuracy and 0.8780 macro F1 on 161 held-out records. Human-reviewed relevance classification reaches 68.33% accuracy on 60 Kandy profile-place pairs; an extension candidate reaches 60.00% on a separate single-reviewer set and remains inactive. These results support bounded component findings rather than overall system accuracy. The study contributes an integrated implementation and an analysis of the evidence needed to distinguish working features, model performance and real travel outcomes.

Keywords— tourism recommendation, landmark recognition, knowledge graphs, itinerary planning, route hazard classification, Sri Lanka

I Introduction

A traveller selecting a destination must also decide where to stay, which attractions fit their interests, how long movement between those attractions will take, and which transport options satisfy the trip's constraints. These decisions are interdependent. A hotel recommendation changes the destination of a vehicle request; replacing an attraction changes the available travel time; identifying a landmark in a photograph creates a need for reliable contextual information. A useful planning application must preserve these relationships while communicating uncertainty in its data and predictions.

Tourism recommender research already considers personalization, mobile context and route planning [1]. The tourist trip design problem formalizes the selection and ordering of points of interest under time and other constraints [2]. We therefore do not claim that personalized tourism planning or the combination of machine learning and routing is new. The research question addressed here is how an integrated Sri Lankan tourism prototype can connect complementary recommendation components while keeping their decision rules and evidential limits explicit.

CeylonGo provides four linked capabilities. Users can identify a supported landmark from an image and ask questions about its stored metadata. Natural language requests retrieve hotels and activities from a graph. A bounded itinerary planner selects and schedules attractions using interests, location and available time. A separate hazard classifier enriches route analysis and supports vehicle selection using passenger, budget and gradient constraints. These components share an application interface, but they do not form a single end-to-end trained model.

The work makes three contributions. First, it describes the implemented data flow connecting the four components, including coordinate based transfer of hotel destinations and ordered itinerary legs. Second, it specifies the separation between learned predictions, deterministic rules, retrieved evidence and optional generated explanations. Third, it reports component evaluation results with their actual units of analysis, exposing generalization and deployment gaps that an aggregate accuracy figure would conceal. The contribution is a system implementation and preliminary component study; superiority over existing travel applications has not been established.

The current geographical scope varies by component. The itinerary catalogue contains 20 Kandy, 10 Matale and 10 Nuwara Eliya points of interest. Landmark recognition covers 17 named classes, including sites outside this region. Hotel, activity and route datasets are separate resources. Their combined presence does not establish complete national coverage or consistent verification across every record.

II Related Work

A Tourism recommendation and trip design

Gavalas et al. describe mobile tourism recommendation as a problem involving user modelling and contextual information [1]. Their trip design survey discusses ordered visits that respect attraction attributes and traveller constraints [2]. This distinction is useful for CeylonGo: a high relevance score alone does not show that a sequence can be completed within a day. Our architecture therefore separates candidate relevance from scheduling and preserves time feasibility checks when a user changes selected stops.

The present prototype uses a small catalogue and bounded search rather than attempting unrestricted national optimization. Its contribution must be assessed through constraint satisfaction, recommendation relevance and usable integration. A future comparative study should evaluate a nearest-first heuristic, a preference-only ranking and the complete planner under the same candidate set, time budget and travel matrix. Without that experiment, the use

Commented [1]: Before submission, add the approved author order, affiliations, emails and supervisor details. Confirm the official title and target venue template. This manuscript uses the local working-tree evidence inspected on 28 September 2026, including uncommitted work; it is not a claim that every branch has been integrated.

of a genetic algorithm is an implementation choice rather than evidence of better routes.

B Landmark recognition

MobileNetV2 uses inverted residual structures and linear bottlenecks to support efficient vision models [3]. Support vector machines provide a separate discriminative classifier that can operate on extracted feature vectors [4]. CeylonGo's landmark experiment combines these approaches by applying an SVM to pooled features from a fine tuned MobileNetV2 network. It also records comparisons with a custom convolutional network and a vision transformer.

The Google Landmarks Dataset v2 study emphasizes the challenges of instance recognition, including out-of-domain photographs, class imbalance and substantial within-class variation [5]. Our 17-class evaluation addresses a much narrower closed-set task. Recognizing a known landmark in a validation image does not establish reliable rejection of an unrelated building or a photograph outside the supported catalogue. An independent image collection and explicit unknown-class evaluation remain necessary.

C Retrieval and generated explanations

Retrieval-augmented generation combines external information with language generation [6]. GraphRAG further uses graph structures and community summaries for query-focused summarization over document collections [7]. CeylonGo uses more limited mechanisms: structured Neo4j queries retrieve hotel and activity records, while landmark metadata is inserted into an optional language model prompt. The safety module uses graph retrieval and deterministic explanation templates. It does not implement the community summarization pipeline described in [7].

We use the term graph-backed retrieval for these implemented operations. Supplying retrieved records can constrain an explanation, but a prompt alone does not prove that generated claims are correct. The proposed evaluation therefore separates retrieval validity, preference extraction and answer faithfulness. None of those measures can be inferred from the number of records stored in the graph.

D Hazard classification and evaluation

Gradient boosting and random forests are established ensemble learning methods [8], [9]. CeylonGo compares them with logistic regression and a decision tree for three-class historical hazard severity classification. The relevant deployment question concerns transfer across routes, so records sharing a route code are grouped during evaluation. Preprocessing and model fitting are composed into scikit-learn pipelines [10]. The predicted class describes the dataset's severity target; neither its label nor its confidence score directly measures the probability of a future accident.

III System Design and Methodology

A Architecture and component boundaries

The client is implemented with React Native and Expo, with web-specific map components where needed. An Express backend coordinates requests, validates inputs and connects the application to Python services, Neo4j and MongoDB itinerary persistence. The Flask service supports itinerary planning and landmark inference. Route hazard prediction is invoked through a bounded Python subprocess. Optional external dependencies provide road routing, geocoding, weather context, hotel rates and language generation.

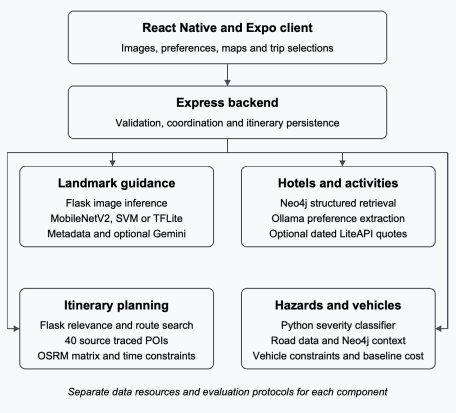


Fig. 1. Implemented component architecture. Connections summarize service responsibilities; they do not imply one shared trained model or one uniformly verified dataset.

Integration preserves concrete trip entities. A chosen hotel's coordinates are carried into vehicle planning, avoiding reliance on a loosely matched destination name. Itinerary safety analysis operates over ordered legs. The planner exposes selected stop identities and planning options so that replacement and schedule rechecking can retain accepted choices. Model inputs, service availability and price sources are kept distinguishable in responses. These are implementation properties; their effect on user effort has not yet been measured.

B Data resources and provenance

Table I summarizes the inspected resources. Counts refer to local files or recorded notebook outputs, not to a census of real-world tourism services. The 40-place itinerary catalogue retains source identifiers, URLs, district labels and coordinates. Its visit durations are planning assumptions, ranging from 60 to 240 minutes according to site type. Source tracing of a place's identity does not verify its current opening hours, entry prices or accessibility.

TABLE I. DATA RESOURCES AND EVALUATION SCOPE

Resource	Size	Scope and qualification
Landmark images	5,500	Notebook records; 17 classes; 4,400 training and 1,100 validation images
Itinerary catalogue	40 POIs	Source traced Central Province records; visit times are assumptions
Relevance v1	240 + 60	Weak training pairs and human-reviewed held-out Kandy pairs
Relevance v2	680 + 60	Weak training pairs and separately evaluated single-reviewer extension
Hazard training	598 rows	22 route groups; source severity mapped to three classes

Resource	Size	Scope and qualification
Roads and vehicles	1,000 / 200	Route-family attributes and vehicle baseline specifications
Hotels and activities	1,830 / 2,000	Local graph inputs; not independently audited inventories

The earlier 1,000-row synthetic/prototype attraction file is retained separately and is not used as evidence of verified national coverage. The hotel CSV is named as an SLTDA master dataset, but the filename alone does not verify every field's provenance. The activity table and baseline vehicle prices also require source and freshness review before broader claims are made. OpenStreetMap attribution and licence requirements apply to relevant data [12].

C Visual landmark recognition and contextual guidance

The notebook resizes images to 224 by 224 pixels and uses a fixed seed of 42 for an 80/20 training-validation split. It records 5,500 images across 17 classes, giving 4,400 training images and 1,100 validation images. Four alternatives are evaluated: a custom CNN, fine tuned MobileNetV2, a vision transformer and deep features with an SVM. MobileNetV2 training includes a classification-head stage and subsequent fine tuning from layer 100 onward, with early stopping and learning-rate reduction.

For the hybrid classifier, global average pooling produces a 1,280-dimensional feature vector. A StandardScaler is fitted to training features and applied to validation features. An RBF-kernel SVM uses C = 1.0 and probability estimation. The notebook exports the scaler and classifier and also converts the neural classifier to TensorFlow Lite. Validation data are used in training decisions, so subsequent scores on those images remain validation results rather than an untouched final test.

The image API returns a supported class and associated metadata, including historical descriptions and visiting information. It applies a 40% confidence threshold below which a photograph is presented as unrecognized. This threshold is an application heuristic; its unknown-image rejection rate has not been evaluated. An optional Gemini request receives landmark metadata and recent dialogue context. When unavailable, local keyword rules produce responses. The metadata and generated responses require factual freshness checks, particularly for prices and opening hours.

A deployment discrepancy is material to interpretation. The notebook extracts SVM features from the fine tuned network, whereas the inspected API constructs an ImageNet-weighted MobileNetV2 feature extractor. Those feature spaces must not be assumed interchangeable. The API also contains fixed accuracy strings that differ from the notebook outputs. Consequently, this paper reports the notebook experiment separately and makes no accuracy claim for the deployed landmark endpoint.

D Hotel and activity retrieval

The hotel graph stores hotels, districts and administrative areas. Activities are connected to districts and to hotels in the same district. Although the relationship is named NEAR_ACTIVITY, its construction uses district equality rather than a measured distance threshold. It therefore represents district association and must not be interpreted as verified walking or driving proximity.

A natural language request is passed to an Ollama-backed extraction step using graph schema values. Parsed fields are validated against the available schema. If extraction fails, deterministic validation and extraction rules are used. The graph query can filter district, hotel category, grade, food type, activity category, price level and suitability. It returns up to ten hotel packages, each with up to five distinct activities. These bounded results are provided to a language model to request a structured day-wise plan. The prompt instructs it to use retrieved hotels and activities, but answer faithfulness remains an empirical question.

Optional LiteAPI enrichment searches around hotel coordinates and checks normalized names before requesting dated stay rates. Unavailable rates are exposed as unavailable. The vehicle component uses dataset baseline pricing; a hotel quote and a vehicle baseline therefore have different evidential status. A combined display is a planning estimate rather than a confirmed booking price. No hotel recommendation accuracy, retrieval precision or booking-success percentage is asserted here.

E Preference based itinerary generation

The planner receives interests, a starting coordinate, a search radius and an available time budget. Seven supported interest categories are adventure, city, culture, history, nature, religion and wildlife. Candidate filtering uses the source traced catalogue, radius constraints and exclusions. An observed rating below 3.9 is screened out, while an unknown rating remains eligible. Rating screening is a deterministic rule, not a learned quality prediction.

The default relevance artifact is a version-1 linear SVM. Its inputs are user interests and POI tags, expanded into transparent category and interaction features. Identity fields, review labels and precomputed target fields are excluded. The classifier gates candidate eligibility; it does not directly predict complete-trip satisfaction. When the artifact cannot be loaded or its hash does not match, the engine reports an explicit content based fallback. The version-2 candidate is separately versioned and is not the default runtime model.

Eligible candidates are ranked using 70% normalized spatial proximity and 30% cosine similarity between interest and tag vectors. These weights are fixed design choices without a reported tuning study. The search considers at most 15 candidates and selects at most eight stops. Initial selection uses a seeded genetic algorithm with a population of 80 and 60 generations. Its fitness rewards relevance and useful visit time while penalizing travel; routes exceeding the time budget receive a very small fitness. Repair operations maintain valid, unique stop identities.

The road mode uses an OSRM driving-time matrix over OpenStreetMap roads [11], [12]. The schedule includes driving, assumed visit durations, waiting for supplied opening windows, a break and an optional return leg. A user-adjustable travel buffer defaults to 25%. Final ordering uses a directed travel-time subset search over selected places. This handles asymmetric travel costs within the bounded plan but does not prove that initial attraction selection is globally optimal.

Opening windows are date-specific user inputs with a source URL. Unknown windows remain unknown, and a visit must finish before a supplied closing time. Stop replacement preserves accepted places, full regeneration searches for an unseen fixed-size subset, and rescheduling keeps selected identities. Exhausted or infeasible requests return controlled errors. In road mode, a routing failure is not silently replaced with a straight-line estimate. Older requests that omit road mode retain the legacy estimate, so evaluations must record which mode was used.

F Route hazard analysis and vehicle selection

The hazard dataset contains 598 rows in 22 route-code groups. Labels are Low, Medium and High, derived from the source Severity Level field. The distribution is 57 Low, 224 Medium and 317 High records. The source severity and route code are excluded from predictor inputs. Numerical features are gradient, elevation, friction, historical occurrence count and a road-data availability indicator. Categorical inputs are terrain, road surface, road width, hazard type and season.

Numerical preprocessing uses median imputation and scaling; categorical preprocessing uses most-frequent imputation and one-hot encoding. The selected gradient boosting classifier has 150 estimators, learning rate 0.05, depth 3 and seed 42. Neo4j retrieval may populate historical count, hazard type and season fields. Current weather is returned as contextual information but is not an input to the deployed classifier. Predicted-class probability is exposed as a model confidence score without a calibration claim.

Road records lack precise traversed-segment geometry. Route matching therefore uses road codes or name evidence and aggregates route-family attributes: maximum gradient, average elevation and friction, and deterministic categorical modes. Where alternatives have distinct usable evidence, the application ranks them by risk class and class probabilities before duration and distance. Where evidence is insufficient, it retains the default analyzed route. This is a comparison among evaluated alternatives, not a guarantee of the safest route.

Vehicle filtering considers the user's budget, passenger count, category and known gradient relative to vehicle gradeability. Ranking uses capability and cost fields such as torque, engine capacity and gradeability margin. The hire estimate is the base charge plus distance multiplied by a per-kilometre rate. These are dataset baselines, not verified current supplier quotes. The rules do not use measured crash outcomes or establish empirical safety superiority of a vehicle model.

IV Evaluation Design

A Separating evaluation questions

We evaluate landmark class prediction, hazard severity prediction and profile-place relevance as separate tasks. They have different targets and sampling units; their percentages must not be averaged into a system accuracy. Hotel retrieval, generated-answer faithfulness, route timing and end-to-end usability require their own studies. The reported numerical results are recorded experiment outputs inspected alongside code and metadata, rather than new full training runs performed for this manuscript.

B Hazard protocol

The hazard workflow first creates a GroupShuffleSplit holdout using route code and seed 42. The 20% parameter applies to route groups, producing 437 development rows across 17 routes and 161 final holdout rows across five routes. Five-fold GroupKFold comparison is restricted to development data. The two leading models then enter ten repeated development-only grouped splits. The architecture is selected by mean validation macro F1, with the mean training-validation gap as a tie-breaker.

Only the selected architecture is evaluated on the final holdout. Afterward, the deployment model is refitted on all 598 rows. Its stored holdout metrics describe the earlier evaluation fit, not an independent test of the final refitted artifact. This order avoids using the final holdout to choose among candidates, although route-family aggregation and source-label quality still limit the external validity of the experiment.

C Relevance protocol

Version 1 uses 240 weak-labelled training pairs from 12 profiles and 60 human-reviewed held-out pairs from three Kandy profiles. Four-fold profile-grouped validation compares candidate models. The stored reviewer evidence records agreement on 53 of 60 pairs, or 88.33%, with unweighted kappa 0.7935. The held-out labels describe judgements about a bounded catalogue, not a study of 60 independent travellers.

Version 2 uses 680 weak-labelled pairs from 17 training profiles. Model selection uses five-fold profile-grouped validation. Its separate extension evaluation contains 60 labels from one reviewer, covering 20 Matale and Nuwara Eliya POIs and three held-out profiles. This set is not pooled with the earlier Kandy results. The stated activation conditions are accuracy at least 0.80, macro F1 at least 0.65, balanced accuracy at least 0.70 and recall at least 0.50 for each represented class.

D Measures and unmeasured outcomes

Classification results include accuracy and F1 measures, with macro F1 giving equal weight to classes and weighted F1 reflecting their support. For future route trials, the implemented evidence export records predicted and observed driving minutes by leg. Mean absolute error should use only legs with observed times, and comparison with Google Maps should use matched legs containing both predictions and the same actual observation. Whole-trip completion also requires all intended stops to be visited within the requested budget. No completed field-trial accuracy or travel-time saving is reported.

V Results and Discussion

A Landmark experiment

Table II reproduces the notebook's recorded validation comparison. The deep-feature SVM has the highest accuracy and weighted F1 among the four models. Its 93.18% validation accuracy exceeds the fine tuned MobileNetV2 classifier's 85.55% by 7.63 percentage points.

That difference is descriptive: no significance test or independent final test is available, and the validation set influenced neural training decisions.

TABLE II. RECORDED LANDMARK VALIDATION RESULTS

Model	Accuracy %	Weighted F1 %
Custom CNN	49.45	47.57
Fine tuned MobileNetV2	85.55	85.54
Vision transformer	42.55	41.55
Deep features and SVM	93.18	93.23

All four rows use the recorded 1,100-image validation set. These figures should not be substituted for field performance. In particular, the SVM feature-extractor discrepancy must be resolved by exporting the exact trained backbone and verifying prediction parity. The quantized TensorFlow Lite path needs a separate post-conversion evaluation with matching preprocessing. Fixed API accuracy labels are not evaluation evidence.

B Route hazard classification

Table III summarizes development cross-validation. Gradient boosting and random forest have similar five-fold macro F1 values. In the ten repeated development splits, gradient boosting records mean macro F1 0.8408 with standard deviation 0.0332, compared with 0.8218 and 0.0357 for random forest. Gradient boosting is selected by the stored decision rule, although its mean training-validation gap, 0.1111, is larger than the random forest gap of 0.0822.

TABLE III. HAZARD MODEL DEVELOPMENT COMPARISON

Model	CV macro F1	Fold SD
Gradient boosting	0.7742	0.0682
Random forest	0.7723	0.0649
Logistic regression	0.7286	0.0379
Decision tree	0.7176	0.0725

The selected architecture records 88.20% accuracy, 0.8780 macro F1, 0.8562 macro precision and 0.9074 macro recall on the 161-row unseen-route holdout. These results indicate agreement with the dataset's severity labels under this split. They do not validate real-time accident prediction, weather-sensitive risk estimation or vehicle safety outcomes. Only five route groups are held out; further external routes and independently checked hazard records are needed before broad deployment claims.

C Human-reviewed relevance

The default version-1 relevance model correctly classifies 41 of 60 Kandy pairs, giving 68.33% accuracy, 0.5196 macro F1 and 0.5652 balanced accuracy. Its recall for class 2 is 2/16, or 12.50%, showing that overall accuracy conceals missed high-relevance pairs. The version-2 extension candidate correctly classifies 36 of 60 pairs, with 60.00% accuracy, 0.5775 macro F1 and 0.6491 balanced accuracy. Since their evaluation sets differ, Table IV is not a controlled before-and-after comparison.

TABLE IV. SEPARATE HUMAN RELEVANCE EVALUATIONS

Evaluation	Correct	Accuracy %	Macro F1
v1 Kandy reviewed set	41/60	68.33	0.5196
v2 extension single reviewer	36/60	60.00	0.5775

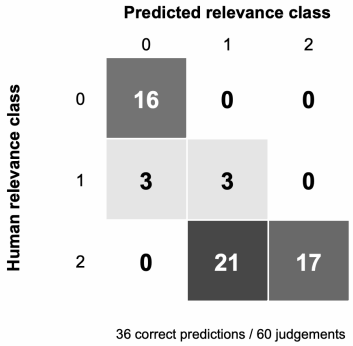


Fig. 2. Recorded v2 extension confusion matrix. Rows are human labels and columns are predictions for 60 profile-place pairs. Labels 0, 1 and 2 denote the ordered relevance classes.

In the extension evaluation, 21 of 38 class-2 examples are predicted as class 1. Class-2 recall is 44.74%, below the activation criterion. Class-1 precision is 12.50%. The candidate fails all four stated activation conditions and remains inactive. This outcome is a substantive finding: expanding weak supervision and reproducing its rules does not establish agreement with human relevance judgements.

The version-2 weak-label cross-validation records 100% accuracy for several non-dummy candidates. Its features directly represent the same interest-tag relationships used to construct weak labels, so this measures rule reproduction. It must not be advertised as human recommendation accuracy. The recorded stratified bootstrap interval for extension accuracy is 0.50 to 0.70; repeated POIs and only three profiles also create dependencies that a row-level interval may not capture. The result is therefore preliminary even within the limited catalogue.

D Integration and remaining evaluation gaps

The implementation links selected hotel coordinates, itinerary stops, route analysis and vehicle inputs. It distinguishes optional language generation from core planning, exposes missing service evidence and supports controlled regeneration. However, an implemented flow is not a measured improvement in planning success. No user-study result, hotel retrieval score or end-to-end latency benchmark is established by the inspected evidence.

The components also propagate different uncertainties. An incorrect landmark prediction can retrieve metadata for the wrong place. A district association can suggest an activity that is geographically inconvenient. Incorrect relevance rejection can remove a desirable stop before route search begins. A feasible static driving schedule can still fail because of traffic, closures or underestimated visit duration. These dependencies motivate evaluation of the complete task in addition to individual classifiers.

VI Limitations and Responsible Use

The study uses a bounded prototype with uneven dataset provenance. Source traced POI identities improve auditability, but visit durations remain assumptions and opening windows are user supplied. Hotel and activity records need a field-level provenance audit. Road characteristics are aggregated by route family rather than exact travelled segments, and historical severity labels require independent validation. Baseline rental rates have no established current provider verification.

The landmark validation set is reused during model development, and the image archive's source licences and near-duplicate controls have not been established here. The deployed SVM preprocessing does not yet match the evaluated feature pipeline. Relevance judgements cover few profiles; the extension has only one reviewer. These limitations restrict both statistical generalization and interpretation of application confidence scores.

OSRM provides road-network estimates rather than a guarantee about future traffic or accessibility. Buffers, road snapping and default visit times can affect feasibility. The prototype does not model all parking, walking, overnight opening windows, live closures or accessibility requirements. A hazard class and a capability-based vehicle ranking should be treated as planning information, not a safety assurance.

Images, trip coordinates and conversational content may be sensitive. Any user study should define consent, retention, access controls and disclosure of external processing before data collection. Human-subject approval, participant consent and institutional endorsement are not asserted in this study. Generated answers require checking against current sources, and any release of datasets must respect source licences and redistribution conditions.

VII Conclusion and Future Work

CeylonGo connects visual landmark guidance, graph-backed hotel and activity retrieval, bounded itinerary generation and route-aware vehicle selection in one tourism prototype. The architecture shows how learned and deterministic components can exchange trip context while exposing missing evidence. Recorded component results are encouraging for some tasks, particularly the bounded hazard experiment, but they do not support a single claim of reliable end-to-end travel planning.

The next evaluation should first align landmark training and inference artifacts, verify post-conversion predictions and introduce an independent, duplicate-controlled test set with unknown landmarks. Relevance modelling needs more independent profiles and reviewers, especially examples distinguishing partial from high relevance. A frozen protocol should compare the default model, an explicit rule baseline and new candidates on the same untouched data before changing runtime activation.

Hotel retrieval should be assessed using labelled user requests and geographic distance checks, followed by extraction and answer-faithfulness measures. Route trials should record actual driving and visit times across multiple dates and conditions. A controlled usability study can then measure task completion, corrections and planning effort. Together, these studies would test whether component integration produces practical benefits beyond the current implementation and preliminary classifier results.

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